SOLID Principles Code Review: Parking Garage System

Principle	Class	Line(s) or Element	Smell / Observation
Single Responsibility Principle (SRP)	`ParkingGarage`	`fee(...)`, `checkout(...)`, `cancel(...)`, `dailyReport()`, `-logger`	Mixes multiple responsibilities: business fee calculations, state/workflow management, report formatting, and direct file infrastructure logging. Multiple reasons to change.
Open-Closed Principle (OCP)	`ParkingGarage`	`fee(Vehicle vehicle, hours : int)`	Calculating fees directly inside the garage based on vehicle types requires modifying `ParkingGarage` every time a new vehicle category or pricing rule is added.
Liskov Substitution Principle (LSP)	`CashPayment`	`refund(amount : double)` *(Note: raises `UnsupportedException`)*	`CashPayment` breaks its parent contract by throwing an exception instead of supporting refunds, making it non-substitutable where a generic `PaymentMethod` is expected.
Interface Segregation Principle (ISP)	`PaymentMethod`	`refund(amount : double)`	Forces cash-based payment methods to implement operations they do not support, leading to forced exception-throwing or unsegregated contracts.
Dependency Inversion Principle (DIP)	`ParkingGarage`	`-logger : FileLogger`	The high-level `ParkingGarage` depends directly on a low-level concrete implementation (`FileLogger`) instead of depending on an abstraction (e.g., a `Logger` interface).